Technical Design Document (TDD): Cold-Chain Logistics AI-Assistant

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Status: Architecture Review

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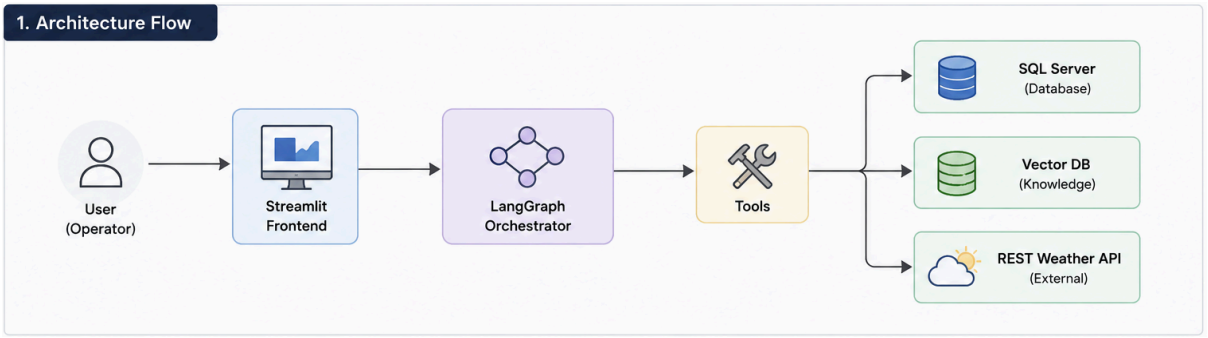
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1. System Architecture (HLD)

The Cold-Chain Logistics AI-Assistant is a decoupled, three-tier enterprise platform designed to enable control tower operators to query fleet telemetry, environmental conditions, and compliance SOPs via a natural language interface.

1.1 Architectural Overview

The system relies on a strictly stateless presentation layer, a state-machine-driven orchestration engine, and a hardened data access layer.



1.2 Component Responsibilities

- **Presentation Tier:** Stateless Streamlit application. Manages UI session state only; all heavy reasoning is offloaded to the Orchestration Tier.
- **Orchestration Engine:** A LangGraph ReAct agent. This component is the "brain." It manages the execution lifecycle, including intent deconstruction, multi-step tool selection, and synthesis of tool outputs into human-readable responses.
- **Data & Integration Tier:** Decoupled backend systems. The system strictly isolates the LLM/Agent from the raw underlying data by utilizing database views and read-only role wrappers.

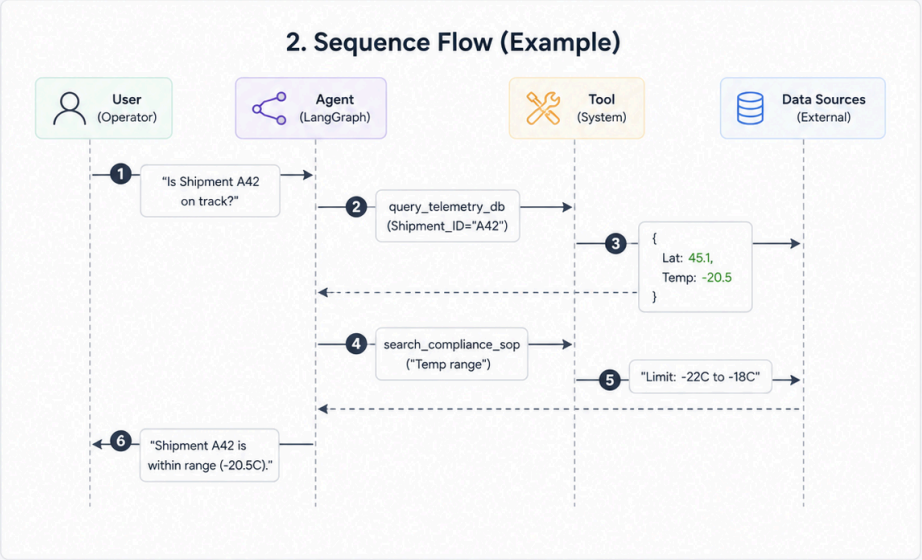
2. Agent Orchestration (LLD)

The orchestration tier ensures deterministic behavior within a probabilistic framework by utilizing a state-machine architecture.

2.1 State Management & Execution Loop

The agent utilizes a LangGraph state object to maintain context throughout the interaction.

1. **Ingestion & Parsing:** The agent receives natural language input and identifies the specific domain (Telemetry vs. SOP vs. Weather).
2. **Reasoning Loop:** The agent evaluates the current graph state. If the intent is ambiguous, it requests clarification. If actionable, it dispatches to the optimal tool.
3. **Tool Execution:** The agent invokes isolated worker functions, ensuring no side effects occur outside the tool's specific scope.
4. **Synthesis:** Raw tool output (e.g., raw JSON, SQL results) is synthesized by the LLM into a cited, context-aware answer.



2.2 Resiliency & Fallbacks

- **Graceful Degradation:** If external API calls (e.g., weather) fail, the agent will report stale/missing data rather than hallucinating status.
- **Self-Correction:** If a SQL query returns an error, the agent logs the error and attempts a simplified query iteration before escalating.

3. Data & Security Model (LLD)

Security is implemented at the physical database layer to mitigate prompt injection risks.

3.1 Telemetry Security (VW_ACTIVE_FLEET)

The agent interacts solely with a read-only SQL view, `VW_ACTIVE_FLEET`.

- **Fields Exposed:** Shipment_ID, Latitude, Longitude, Cargo_Type, Current_Temperature.

- **Security Principle:** By exposing only a flat, view-based schema, we physically prevent the agent from executing DROP, UPDATE, or DELETE commands.

3.2 Role-Based Access Control (RBAC)

Database connections are established using the `USR_FDE_RO` service account. This account lacks permissions to write, modify schema, or access tables outside the specified view.

3.3 Immutable Auditing

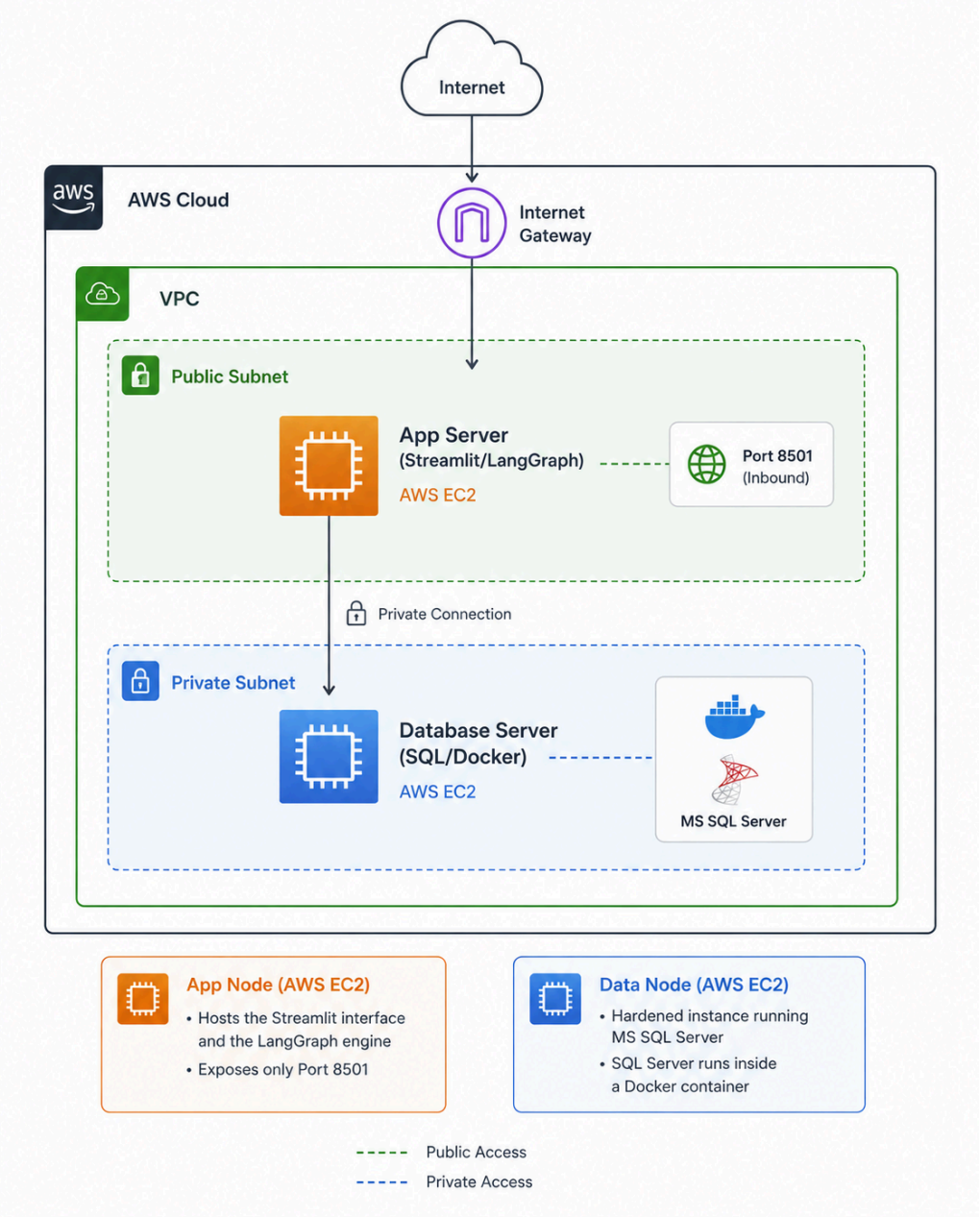
Every decision cycle is committed to `AgentAuditLog` (append-only).

- **Log Entry:** {Timestamp, User_Prompt, Tool_Invoked, Tool_Raw_Output, LLM_Decision}.

4. Deployment Strategy

The system utilizes an infrastructure-as-code (IaC) approach.

4.1 Infrastructure Topology



4.2 Network Perimeter Security

1. Ingress (Public): Only Port 8501 (App Node) is accessible.
2. Internal Communication: Port 1433 (SQL) is restricted via Security Group rules to accept traffic only from the private IP of the App Node. The database is invisible to the public internet.